

E-Commerce



Problem Statement: Customer Behavior & Sales Performance Analysis

Business Scenario

A fast-growing e-commerce company has seen a surge in orders and customers, but management is facing challenges in understanding customer behavior, product performance, and operational efficiency.

They want to identify high-value customers, improve customer retention, optimize marketing spend, and understand how delivery performance impacts customer satisfaction.

As a Data Analyst, your task is to analyze the available data and provide actionable business recommendations.

Challenge

Using the provided e-commerce datasets, analyze customer behavior, sales performance, and operational trends to help the company make data-driven business decisions.

Key Business Questions

1. Which product categories generate the highest revenue?
 2. Which categories are underperforming?
 3. Who are the most valuable customers?
 4. Can customers be segmented based on purchasing behavior?
 5. How do delivery delays impact customer reviews and ratings?
 6. Are there any seasonal or monthly sales trends?
 7. Which categories and regions should receive additional marketing investment?
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Dataset

Brazilian E-Commerce Public Dataset by Olist

Includes multiple linked datasets:

- Orders
 - Order Items
 - Customers
 - Payments
 - Reviews
 - Products
 - Sellers
 - Geolocation
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Expected Tasks

- Merge and clean multiple datasets.
 - Perform customer segmentation / Analysis
 - Analyze product and sales performance.
 - Perform at least one correlation analysis.
 - Create visualizations and dashboards.
 - Provide business recommendations.
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Required Deliverables

- Data Cleaning
- Analysis
- Insights

Dashboard

Power BI / Tableau / Streamlit

PPT Presentation (Maximum 5 Slides)

PPT Should Cover

1. Business Understanding
 2. Key Insights
 3. Dashboard Overview
 4. Recommendations
 5. Business Impact
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